

Assignment: Day 5

Network Testing Tools

Q1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Ans: I opened Command Prompt and ran ping `www.youtube.com`. All 4 packets were received with 0% loss.

```
Command Prompt

C:\Users\Student>ping www.youtube.com

Pinging youtube-ui.l.google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=18ms TTL=117
Reply from 142.250.195.78: bytes=32 time=21ms TTL=117
Reply from 142.250.195.78: bytes=32 time=17ms TTL=117
Reply from 142.250.195.78: bytes=32 time=20ms TTL=117

Ping statistics for 142.250.195.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 21ms, Average = 19ms
```

Average response time = 19 ms (minimum 17 ms, maximum 21 ms).

Q2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Ans: I ran tracert `www.flipkart.com`. The trace finished in 9 hops; hops 1 to 8 are the intermediate devices and hop 9 (163.53.247.20) is the Flipkart server itself.

```
Command Prompt

C:\Users\Student>tracert www.flipkart.com

Tracing route to www.flipkart.com [163.53.247.20]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    192.168.1.1
  1  4 ms     3 ms     4 ms     10.22.0.1
  2  6 ms     5 ms     6 ms     103.87.140.1
  3  8 ms     9 ms     8 ms     172.31.250.21
  4  *        *        *        Request timed out.
  5  21 ms    22 ms    21 ms    14.140.210.5
  6  48 ms    51 ms    49 ms    115.113.165.42
  7  50 ms    49 ms    52 ms    182.79.198.106
  8  51 ms    50 ms    50 ms    163.53.247.20

Trace complete.
```

Hop	IP address	Device
1	192.168.1.1	My WiFi router (default gateway)
2	10.22.0.1	Internet service provider's local gateway
3	103.87.140.1	ISP access router
4	172.31.250.21	ISP internal router
5	Request timed out	Router that does not reply to trace requests (path still continues)
6	14.140.210.5	ISP backbone router
7	115.113.165.42	Backbone router (large jump in delay here)
8	182.79.198.106	Router near the Flipkart network

Q3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Ans: I ran nslookup www.zomato.com. The DNS server (my router, 192.168.1.1) returned the answer below.

```
Command Prompt

C:\Users\Student>nslookup www.zomato.com

Server: UnKnown
Address: 192.168.1.1

Non-authoritative answer:
Name: www.zomato.com
Address: 104.18.20.115
```

Result: www.zomato.com = 104.18.20.115

Q4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Ans: I pinged my phone, which was connected to the same WiFi, using its IP address 192.168.1.7. It replied to all 4 packets in about 4 ms, which shows the LAN connection is working:

```
Command Prompt

C:\Users\Student>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:
Reply from 192.168.1.7: bytes=32 time=4ms TTL=64
Reply from 192.168.1.7: bytes=32 time=3ms TTL=64
Reply from 192.168.1.7: bytes=32 time=5ms TTL=64
Reply from 192.168.1.7: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 4ms
```

Then I turned off the phone's WiFi and pinged again. Every request showed *Request timed out* and the result was 100% packet loss:

```
Command Prompt

C:\Users\Student>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

What happens when a device is offline: no reply comes back, so ping keeps showing "Request timed out" (or "Destination host unreachable") and reports 100% loss. This can mean the device is switched off, disconnected from the network, or blocking ping with its firewall.

Q5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Ans:

(a) Website not loading: ping. ping www.youtube.com got 4 replies with 0% loss, so the internet works and the fault would be with the website or browser.

(b) Slow connection: traceroute. In my tracert to www.flipkart.com the delay jumped from 21 ms (hop 6) to 48 ms (hop 7), so the slowness is in the ISP backbone, not my WiFi.

(c) Suspecting DNS issues: nslookup. nslookup www.zomato.com returned 104.18.20.115, so DNS is working; a "can't find" error would point to DNS.